Marine Microbial Biodiversity

Lecture 09

Ocean Habitat

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The Ocean In Numbers

Seawater covers approximately 71% of Earth's surface and 90% of the Earth's biosphere

The ocean contains 97% of Earth's (surface) water

The total volume is approximately 1.34 x 10^9 Km^3

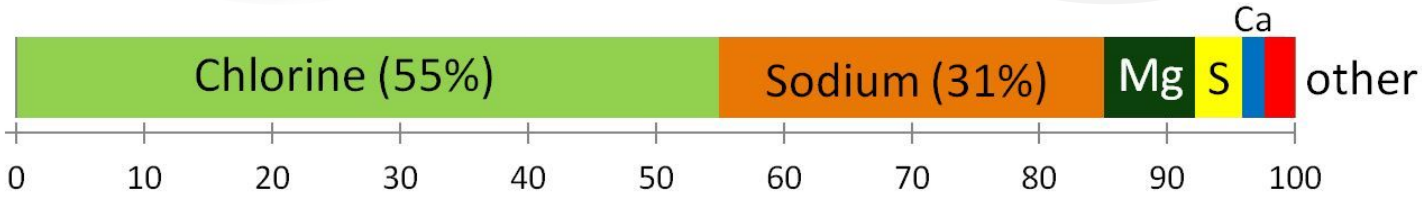
Av. depth of more about 3,682 meters

Av. salinity is around 34.7‰ (parts per thousand or ppt) or 3.4%

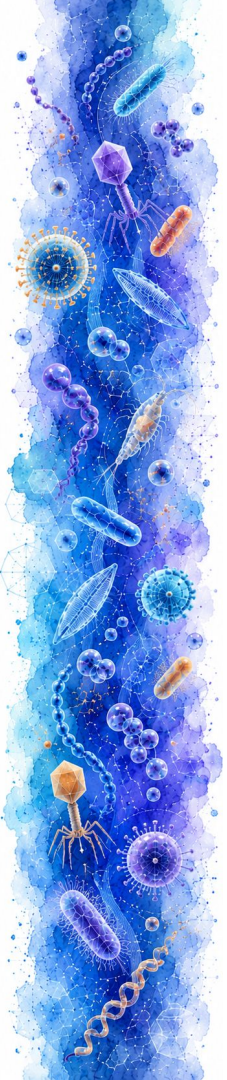
Av. temperature of ocean's surface waters is about 17 °C, deep-sea temperature is within the 1.5 °C

Seawater freezes at around -1.9 °C at atmospheric pressure

Less than 5% of the World Ocean has been explored in any details, and even less of marine sediments and subsurface



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Area And Depth Of Seas And Oceans

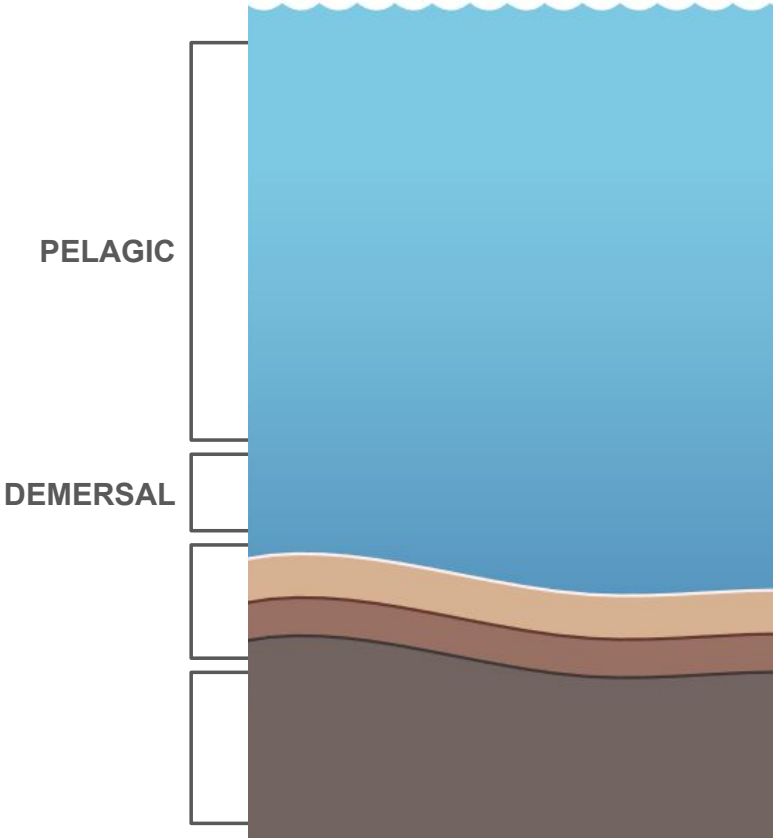
Name	Area Sq km	Average depth m	Greatest known depth m	Place of greatest known depth
Pacific Ocean	155,557,000	4,028	11,033	Mariana Trench
Atlantic Ocean	76,762,000	3,926	9,219	Puerto Rico Trench
Indian Ocean	68,556,000	3,963	7,455	Sunda Trench
Southern Ocean	20,327,000	4,000–5,000	7,235	South Sandwich Trench
Arctic Ocean	14,056,000	1,205	5,625	77°45′N; 175°W
Mediterranean Sea	2,965,800	1,429	4,632	Off Cape Matapan, Greece
Caribbean Sea	2,718,200	2,647	6,946	Off Cayman Islands
South China Sea	2,319,000	1,652	5,016	West of Luzon
Bering Sea	2,291,900	1,547	4,773	Off Buldir Island
Gulf of Mexico	1,592,800	1,486	3,787	Sigsbee Deep
Okhotsk Sea	1,589,700	838	3,658	146°10′E; 46°50′N
East China Sea	1,249,200	188	2,782	25°16′N; 125°E
Hudson Bay	1,232,300	128	183	Near entrance
Japan Sea	1,007,800	1,350	3,742	Central Basin
Andaman Sea	797,700	870	3,777	Off Car Nicobar Island
North Sea	575,200	94	660	Skagerrak
Red Sea	438,000	491	2,211	Off Port Sudan
Baltic Sea	422,200	55	421	Off Gotland

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Ecological Zones In Marine Environment

Pelagic: The pelagic zone encompasses the open ocean, away from the shore and above the seafloor. It is subdivided into different layers based on depth, light availability, and pressure. The pelagic zone can be thought of in terms of an imaginary cylinder or water column that goes from the surface of the sea almost to the bottom. This zone is home to a diverse range of organisms, from microscopic plankton to large migratory fish and marine mammals.

Demersal: The demersal zone refers to the water column near to the seabed (and is significantly affected by), just above the benthic zone, often near coastal and continental shelf areas.

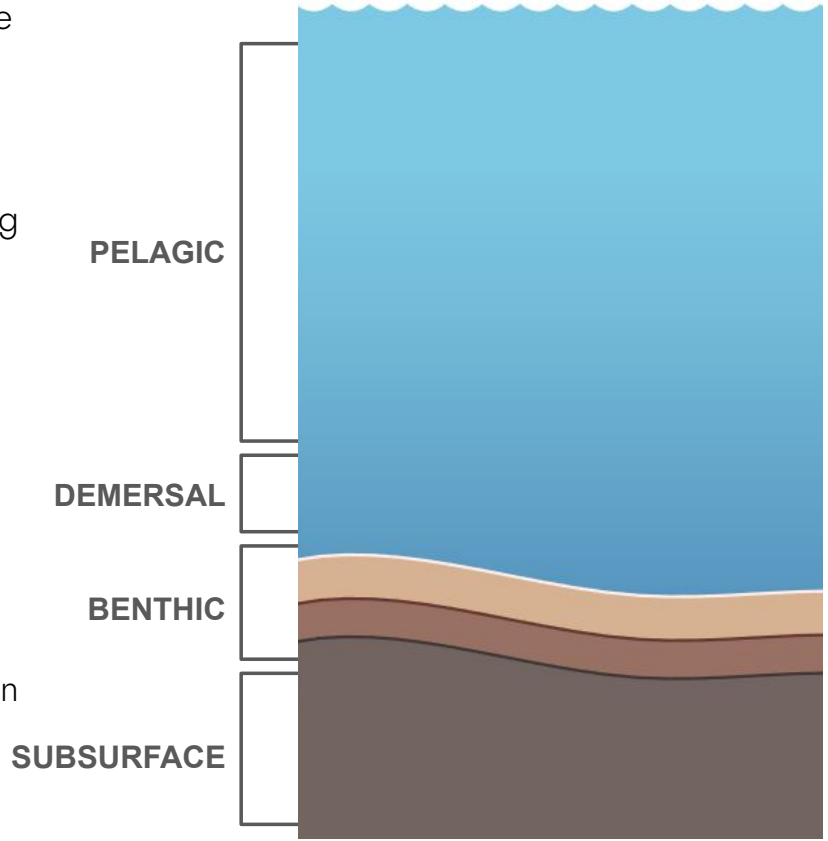


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Ecological Zones In Marine Environment

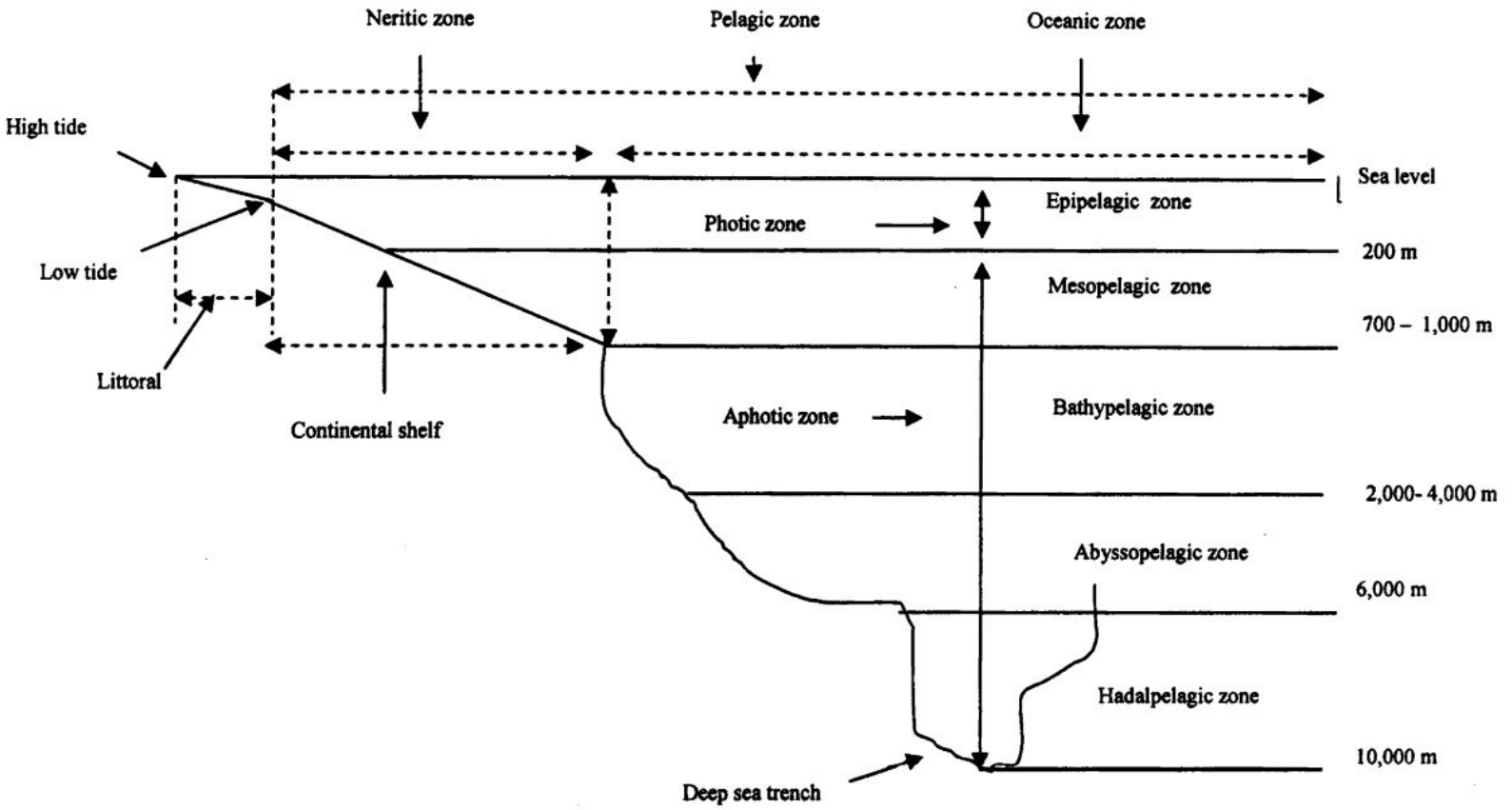
Benthic: The benthic is the ecological region at the lowest level of a body of water including the seafloor, the sediment surface and some sub-surface layers. It refers to environments ranging from shallow coastal areas to the deepest ocean trenches. The superficial layer of water directly above the sediment at the bottom of a body of water represents the benthic boundary layer.

Subsurface: The subsurface is the ecological region below the seafloor. Although the actual starting depth is debated, it is currently set between 0.15 and 1 meters below sea floor (mbsf) and extends to high depths (several kilometers)



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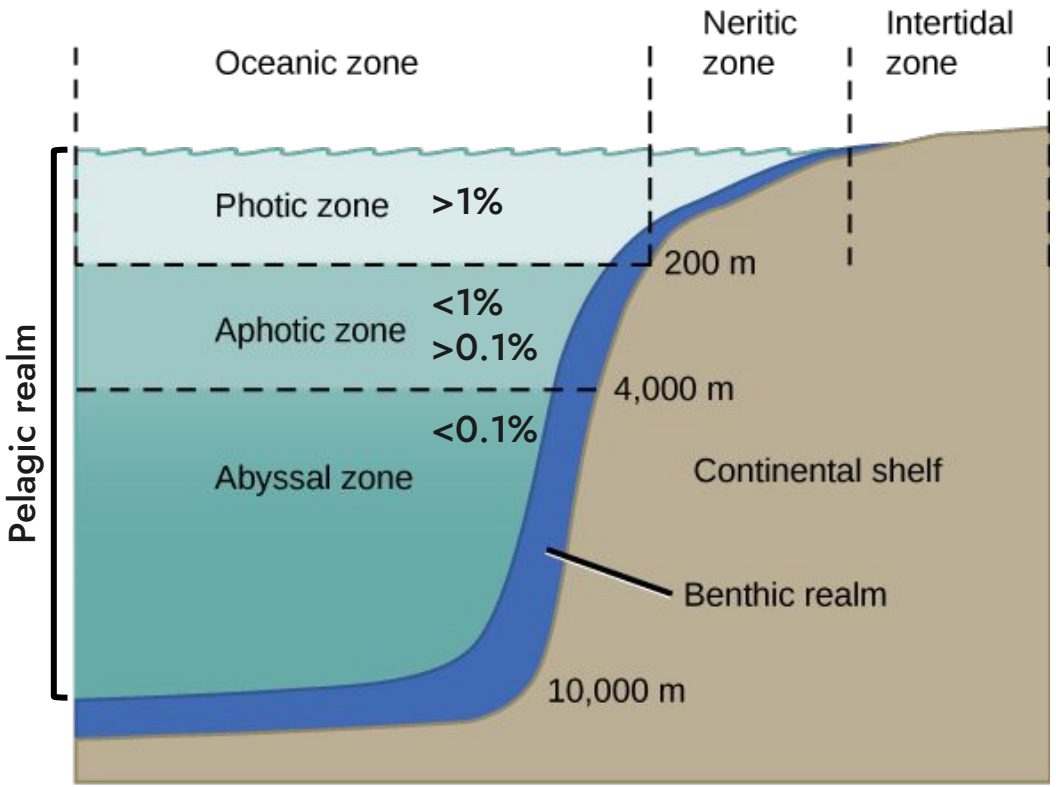
Slicing The Pie: By Light Or By Depth?



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Slicing The Pie: By Light Or By Depth?

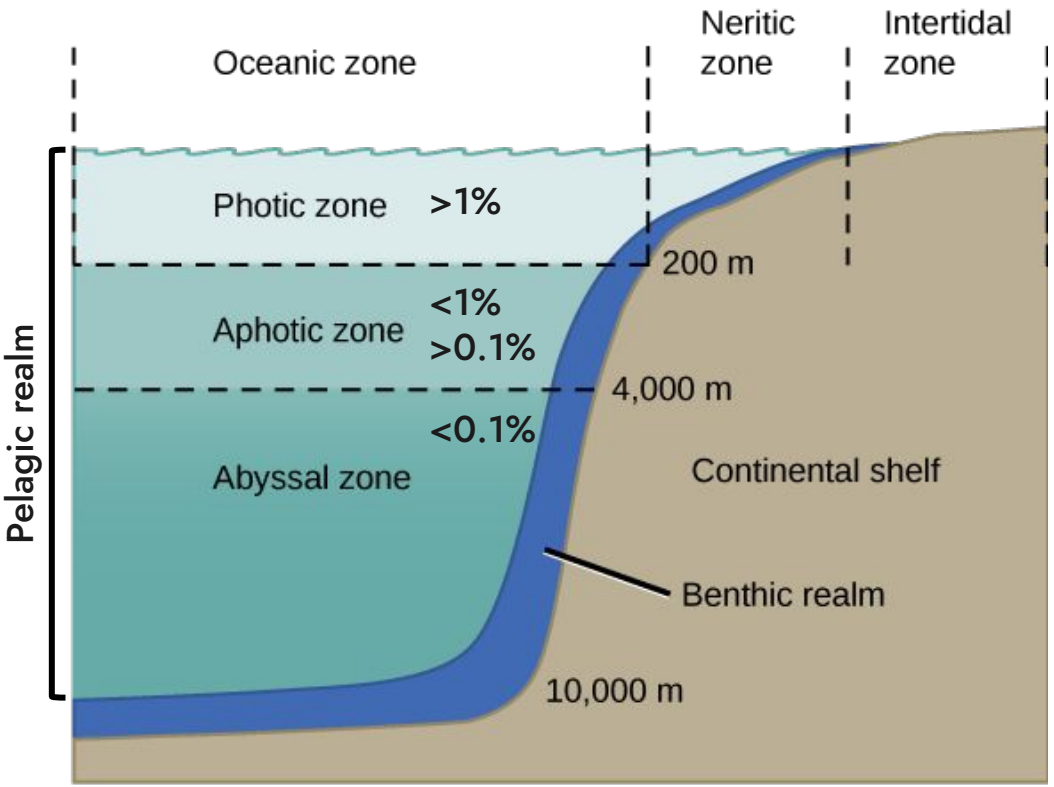
The **photic zone**, or the portion of ocean receiving sunlight, also called the euphotic zone, is defined as the depth where day-light photosynthetically available radiation (PAR) is reduced to max 1% of the value measured just below the sea surface. Typical euphotic depths vary from only a few centimeters in highly turbid eutrophic lakes, to around 200 meters in the open ocean.



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Slicing The Pie: By Light Or By Depth?

The **aphotic zone** is the portion of a lake or ocean where there is **little or no sunlight**. It is formally defined as the depths beyond which **less than 1% of sunlight penetrates**. Often the aphotic zone is further divided in a twilight zone (or disphotic zone, between 1 and 0.1% of sunlight) and the true aphotic or abyssal zone (less than 0.1% of sunlight).



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Slicing The Pie: By Light Or By Depth?

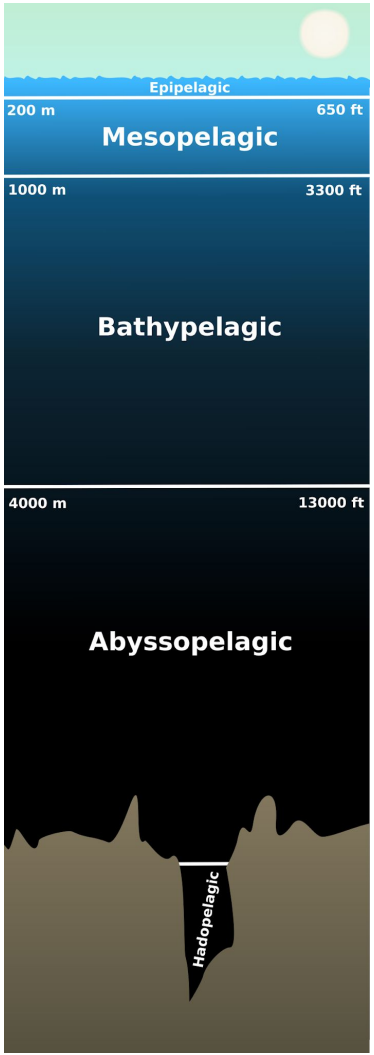
Epipelagic = up to 200 m depth, roughly corresponding to the euphotic zones and the depth at which the continental shelf breaks

Mesopelagic = from 200 m to 1,000 m; temperature varies less than the epipelagic (between 20 and 4°C; is the location of the thermocline

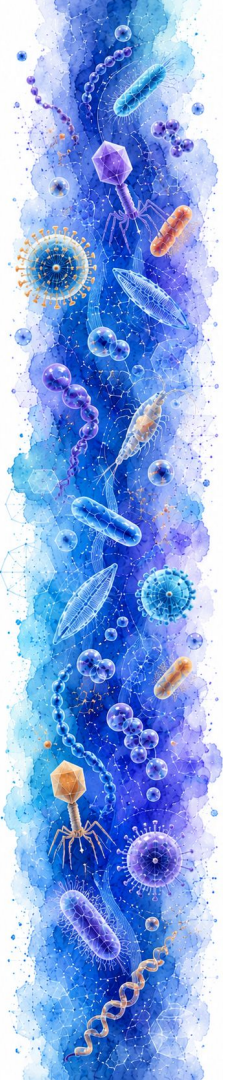
Bathypelagic = from 1,000 m to 4,000 m; also known as midnight zone; average temperature hovers at about 4 °C; majority of the ocean depth is in this zone

Abyssopelagic = from 4,000 m to 6,000 m; temperatures around 2 °C to 3 °C

Hadopelagic = below 6,000 m; also known as the hadal zone and trench zone



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Organisms Classification

Planktonic: populations of free floating organisms in water. Planktons include some bacteria, algae, as well as some protists, not endowed with a motility apparatus

Nektonic: in contrast to planktonic organisms, nektonic organisms are those which, while subject to aquatic flow, also have their own motility apparatus, and hence can also move apart from being carried with the flow of water, e.g., ciliated protozoa, flagellated bacteria, etc…

Periphytic: aquatic microorganism that attached to underwater surfaces. On the basis of the nature of their support, they are defined **epiphytic** if attached to the surfaces of plants, **epizootic** to the surface of animals, **epilithon** when attached to rocks or stones of the aquatic environment, **epixylon** if the community is found on the fallen woods in water bodies, **episammon** if attached to sand grains

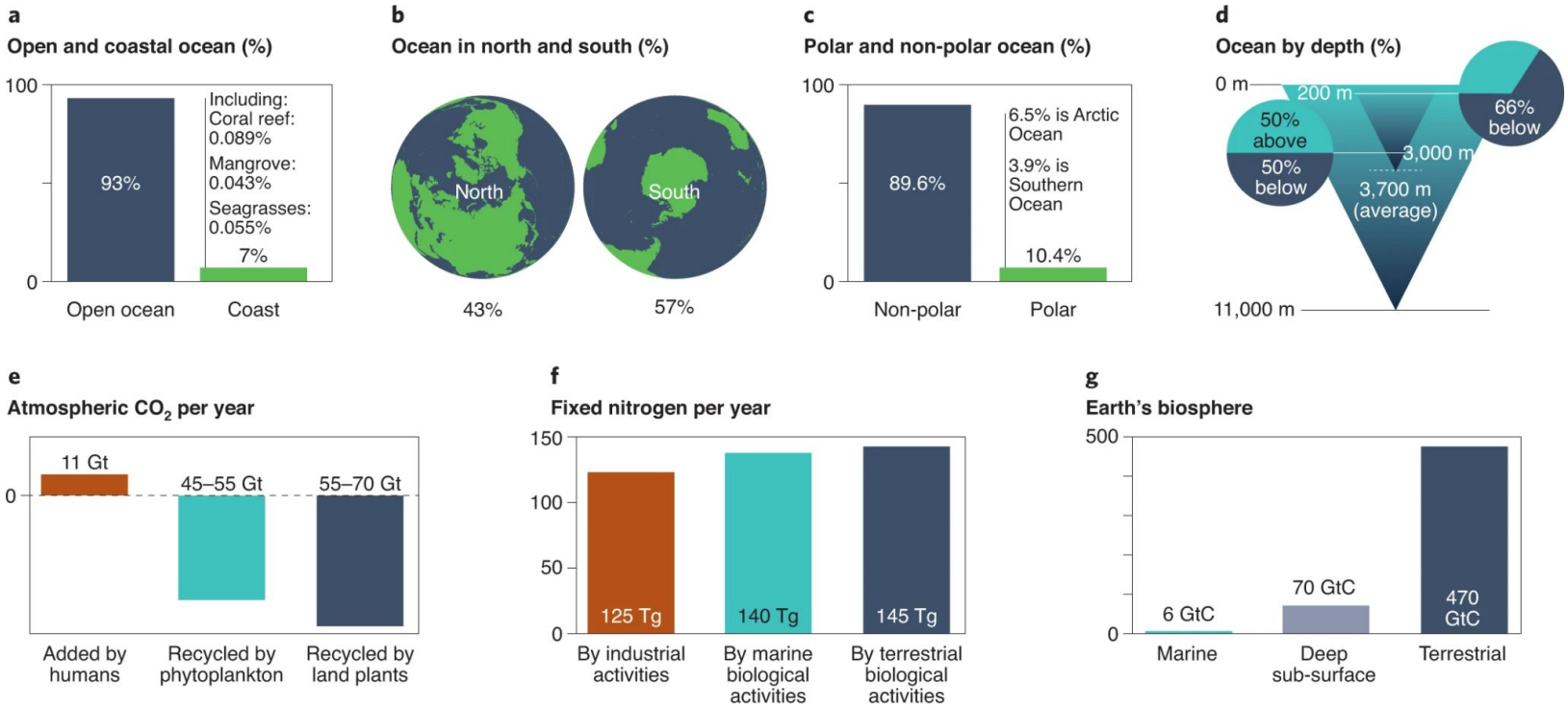
Benthic: organisms inhabiting the bottom sediments of aquatic environments (i.e., bottom or mud-dwelling organisms) constitute the benthos

Neuston: the microorganisms which are found at the surface of an aquatic environment, exactly at the air–water interface, more precisely organisms that float on the top of the water (**epineuston**) or live right under the surface (**hyponeuston**)

Seston Epipelon: sestons are the particulate matter suspended in bodies of water such as lakes and seas, like organic detritus and inorganic material. Microorganisms inhabiting these detritus matters are referred to as seston epipelon.

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Key Features Related To The Ocean Microbiome

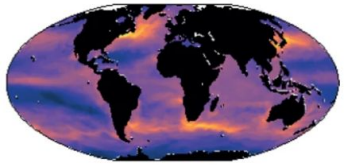


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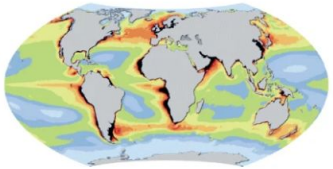
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Key Features Related To The Ocean Microbiome

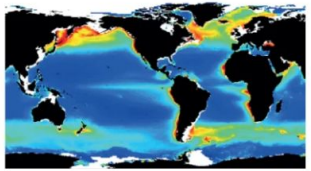
h
Phytoplankton biodiversity



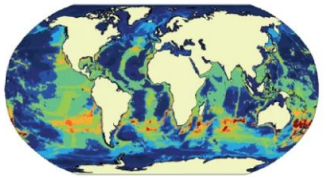
i
Primary productivity



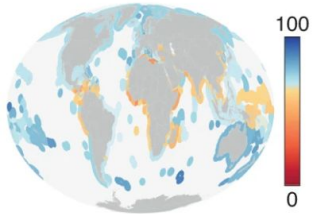
j
Carbon export



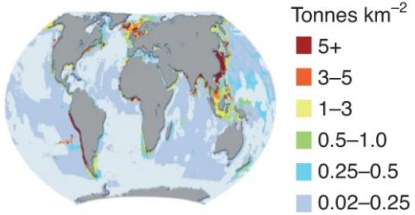
k
Mammalian species richness



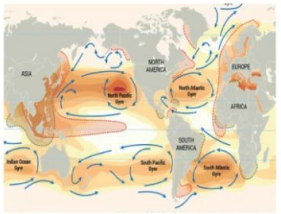
l
Ocean health index



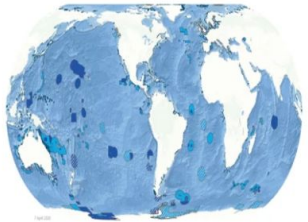
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Fish catch



n
Microplastics



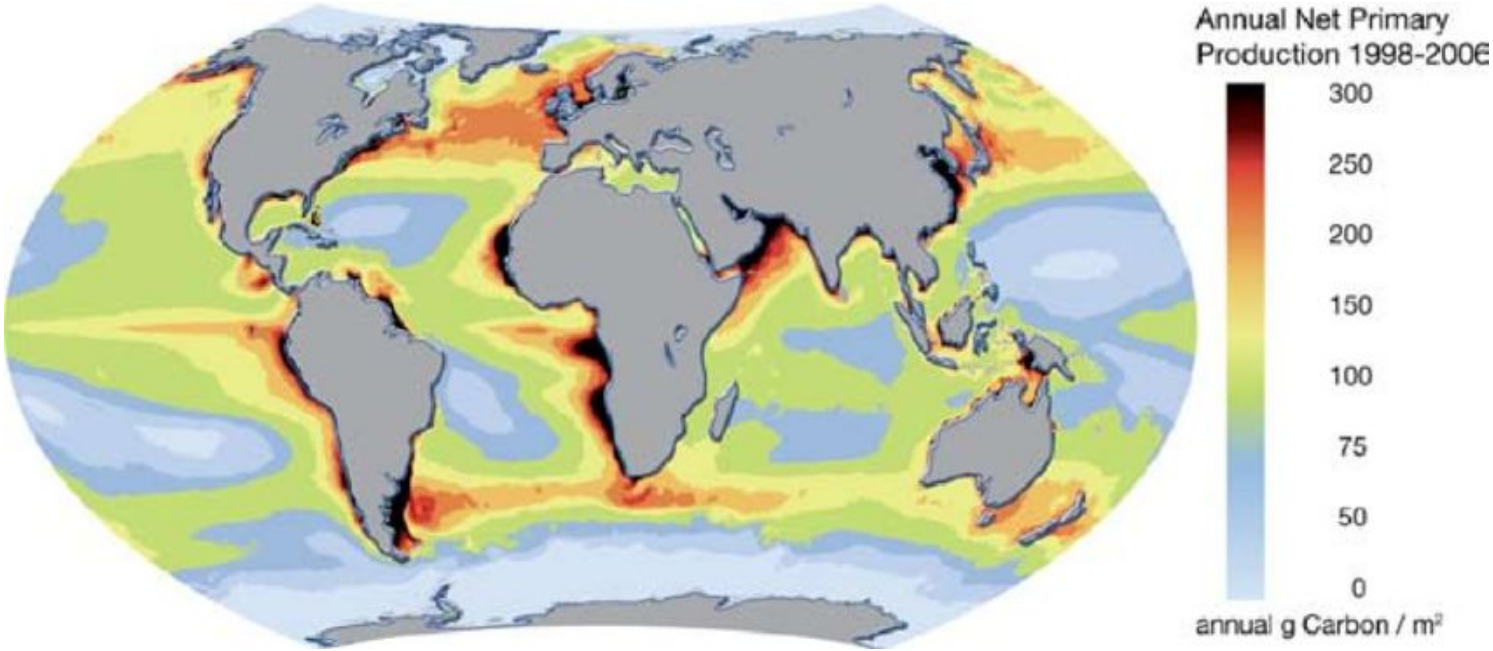
o
Marine protected areas



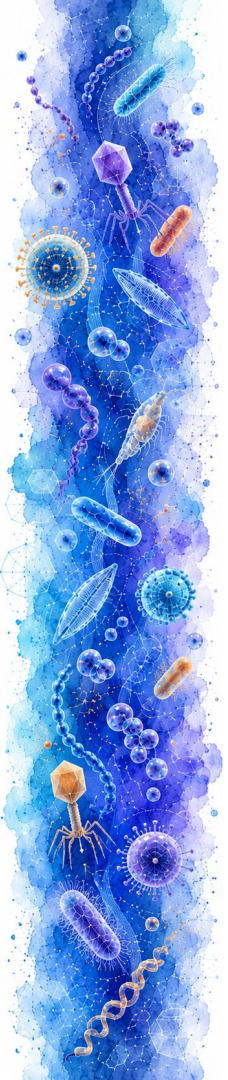
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Surface Primary Production



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Primary production in the Marine Environment

Primary Production (PP) is the synthesis of organic compounds from inorganic precursors, namely CO_2, HCO_3^- and CO. It occurs in the surface oceans and surface sediments within the photic zone, through photosynthesis. Chemosynthesis, while also present in the photic zone in certain areas, is the primary production metabolism in the aphotic zone and subsurface. Despite this, marine biologists commonly associate PP to photosynthesis only. The organisms responsible for primary production are known as primary producers or autotrophs, and form the **base of the food chain**.

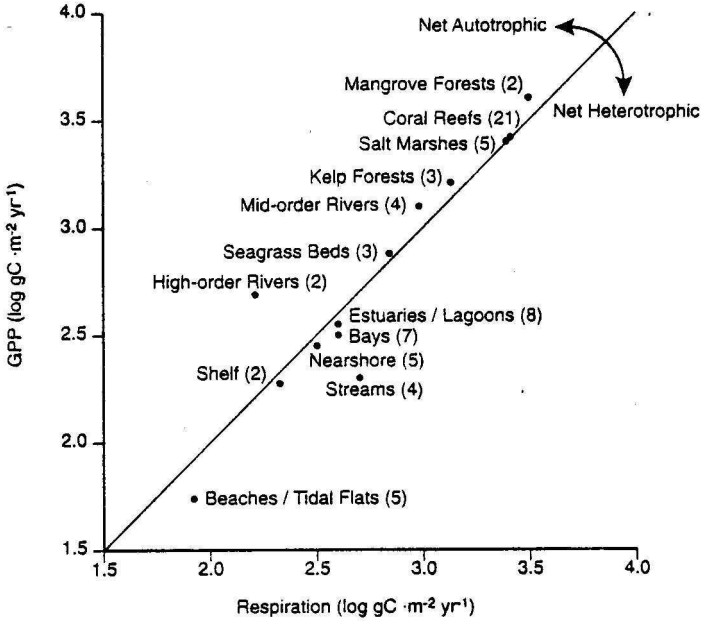
Gross Primary Production (GPP) represents the total amount of organic carbon produced by primary producers in a given length of time. In other words, is the **rate at which primary producers obtain and collect biomass for all energy requirements**. A fraction of this total energy is used by primary producers for **biomass build up**, another fraction is used for **cellular respiration and maintenance** of physiological functions (i.e., "growth respiration" and "maintenance respiration").

Net Primary Production (NPP) is the rate at which all the primary producers produce net useful chemical energy. It can be expressed as the difference between the GPP and the respiration rate, and corresponds to that **energy that is available to the rest of the ecosystem** and can be used by consumers.

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Primary Production VS Respiration

The balance between primary production and respiration, and therefore the **net primary production**, is important to define an ecosystem. When primary production is high, more organic material is produced than consumed, which means there are surplus nutrients available for heterotrophic organisms at higher trophic levels. If respiration outpaces primary production, organic material is depleted faster than it is replenished. This can lead to nutrient limitations, reducing energy flow and potentially destabilizing food webs by limiting resources for organisms dependent on primary producers. In this way we can distinguish net autotrophic and net heterotrophic ecosystems.



Scatter plot of ecosystem gross primary production (GPP) vs respiration

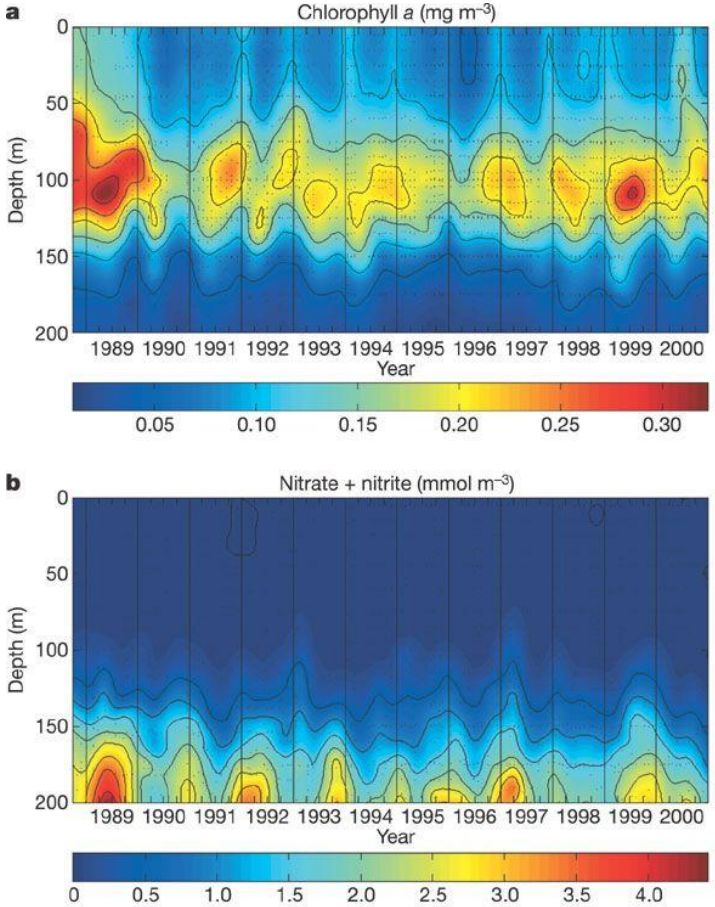
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Factor Limiting Surface Primary Production

The major factor controlling primary production in the marine environment are:

- Availability of **light** (Photosynthetically active radiation - PAR)
- Availability of **nutrients** (N and P)
- Availability of **trace elements** (e.g. Fe, Co)

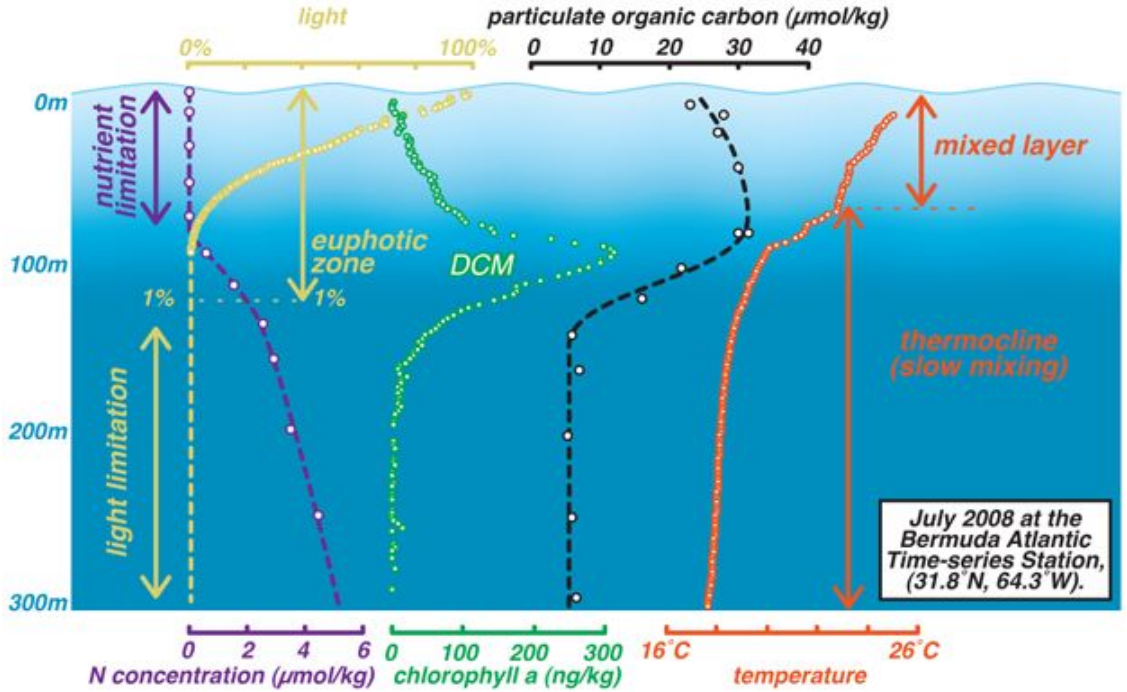
Photosynthesis is limited by the supply of essential nutrients in the sunlight photic zone, and by light in the lower, dark, aphotic zone, within which nutrients are instead abundant. Chemosynthesis is instead limited by the availability of suitable electron acceptor or donor, and or by the direct competition with phototrophs in the photic zone.



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What is the most important nutrient for PP?

Depends on the PP we are referring to, the ecosystem and where we are on the planet. It furthers depends on the organism in question. For example, iron limits photosynthetic PP at high-**latitude** during the summer, whereas in the winter at those latitude light is the limiting factor. Dissolved **silica** might limit diatoms PP, while available **ammonium** might limit Ammonia Oxidizers PP in the bathypelagic zones.

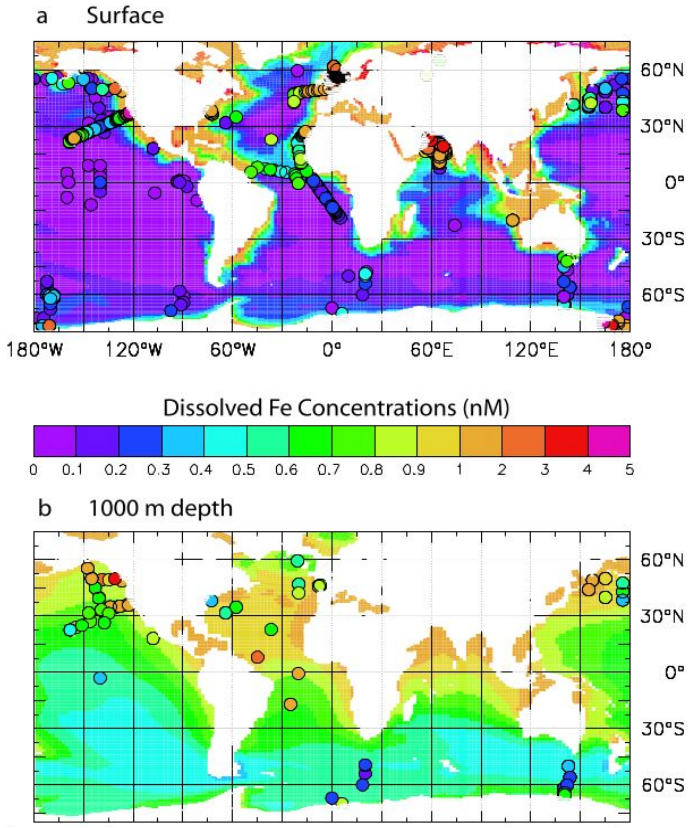


THE MOST IMPORTANT NUTRIENT IS THE LIMITING NUTRIENT

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What is the most important nutrient for PP?

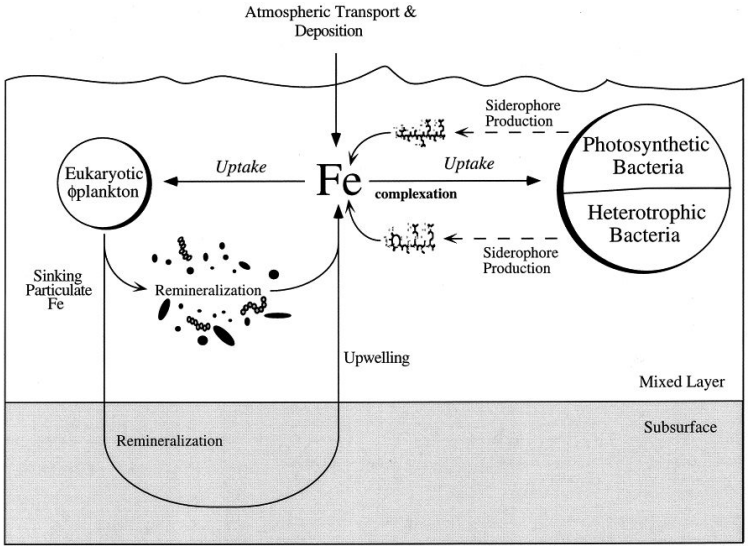
Iron is an important element in biological systems, and it is used to power photosynthesis (e..g, ferroxidine and cytochromes), as a cofactor for nitrogen fixation, nitrate reduction, oxidative phosphorylation, sulfite reduction and other important cellular functions. Iron is a scarce solute in seawater, and the abundance of dissolved oxygen in seawater means that what little iron is present in the ocean is typically in the oxidized ferric iron form. This oxidized Fe^{3+} forms inert iron oxides and hydroxides, all forms not available to microorganism. Fe^{2+}, the reduced ferrous iron form, is more soluble and thus more biological reactive, even if it also oxidizes very rapidly. Upwelling phenomenon is a way to put deep water up and so transport iron.

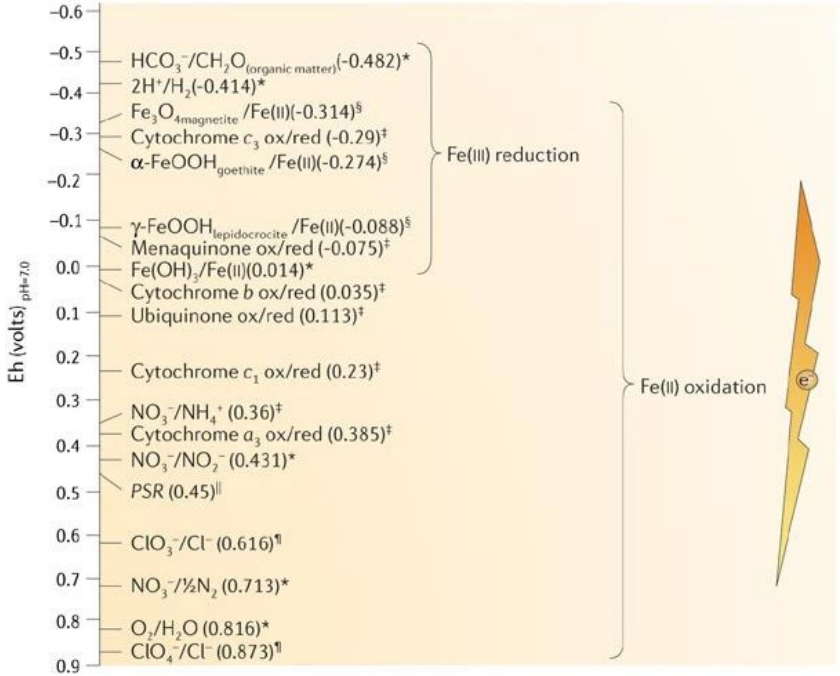


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What is the most important nutrient for PP?

Microbes play a role in recycling iron within the euphotic zone, allowing it to be reused by phytoplankton and other organisms. This "microbial iron pump" helps maintain iron availability in surface waters.





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The Iron Hypothesis: A Geoengineering Idea

"Give me a half ship of iron, and I will give you an ice age."

John Martin, 1988

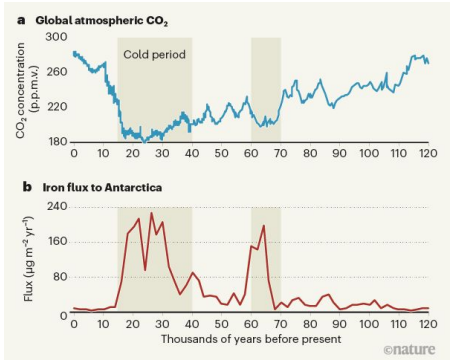


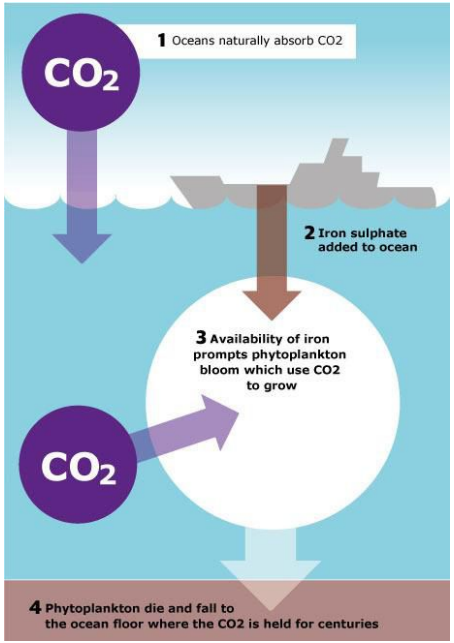
In many parts of the open ocean (e.g., Pacific and Southern Oceans), there are plenty of nutrients like N and P, but very little marine life. These are called HNLC (High-Nutrient, Low-Chlorophyll) zones.

Iron is a crucial micronutrient for phytoplankton (microscopic algae). In HNLC zones, a lack of iron acts as a bottleneck, stopping algae from growing and exploding in massive "bloom".

Like plants on land, phytoplankton absorb CO_2 via photosynthesis. When they die, they sink to the deep ocean floor, effectively locking that carbon away for centuries.

John Martin pointed out that during past ice ages, dust storms carried massive amounts of iron-rich soil into the oceans, naturally fueling this process and dropping global temperatures.



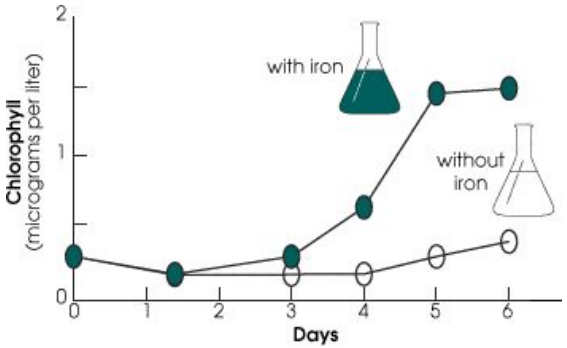


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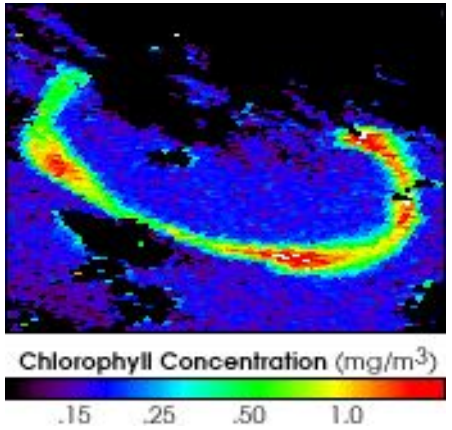


.Early **small-scale** experiments proved that adding Fe does create massive algae blooms.

.Massively altering the food chain could cause toxic algal blooms (like red tides) or starve marine life downstream by using up all the other nutrients (unpredictable **ecosystem shifts**).

.Dead algae sinking to the very bottom of the ocean is not guaranteed. Experiments showed that much of the OM gets eaten by fish or bacteria near the surface, releasing the CO_2 right back into the atmosphere.

.Decomposing algae at depth can deplete oxygen and accidentally trigger the release of CH_4 or N_2O.



.Iron fertilization projects: e.g., **IRONEX (1993)**, **SOIREE (1999)**, **SOFeX program (2002-2007)** involved hundreds of scientists and numerous institutions worldwide.Because of these risks, large-scale ocean iron fertilization is currently restricted by international agreements like the **London Convention on marine pollution**.

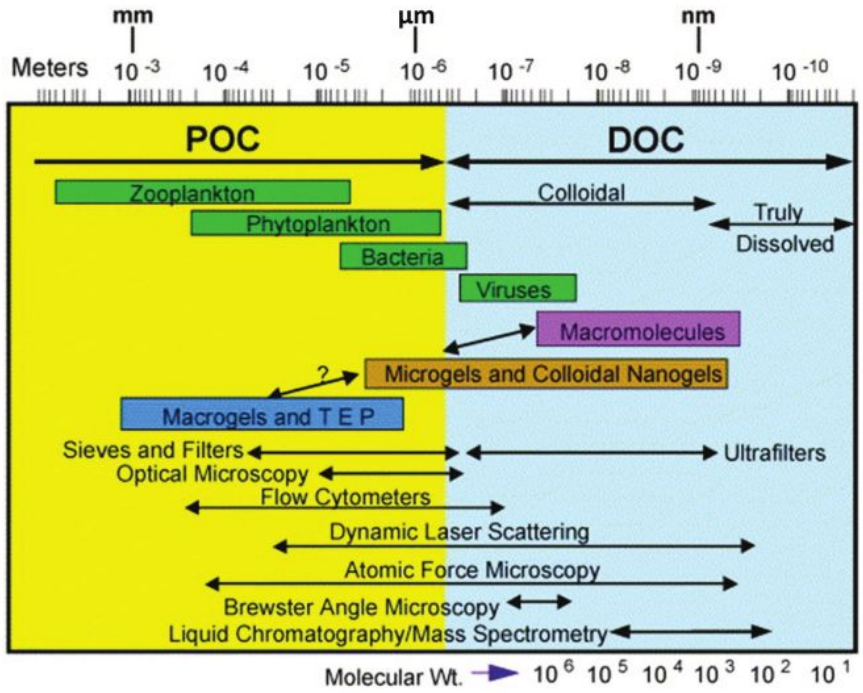
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Primary Production: POC and DOC

Not all the fixed carbon from primary production is converted in biomass. Significant amounts of fixed carbon are released in the environment as Dissolved and Particulate Organic Carbon pools (DOC and POC).

DOC and POC can derive from the direct exudation and excretion of organic chemicals, or can be released during grazing and cell lysis, either due to natural senescence or viral lysis.

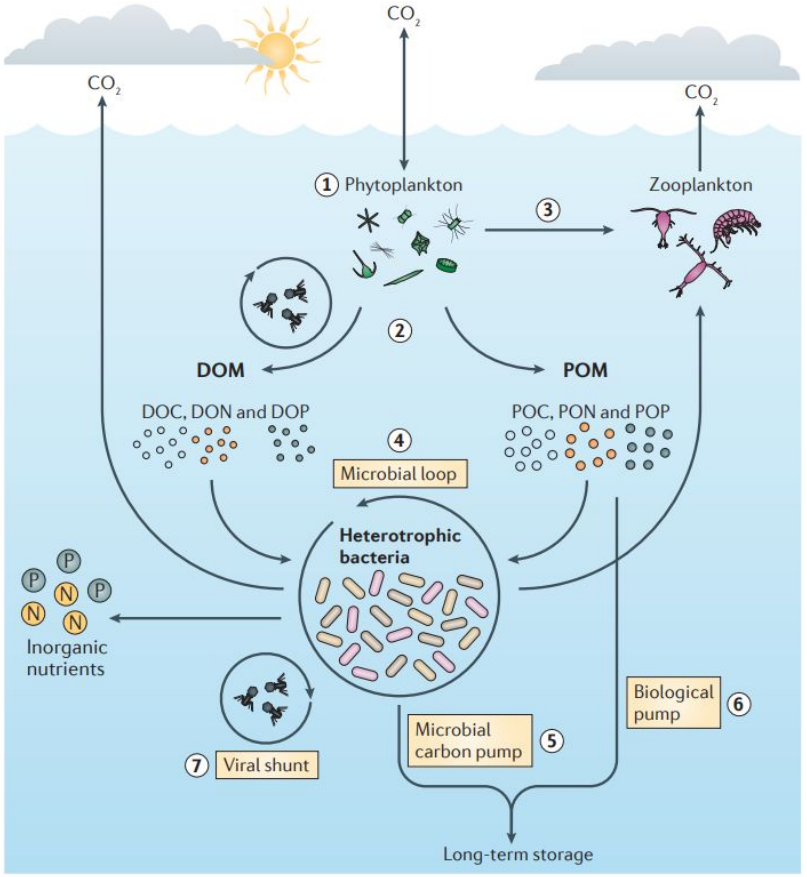
The division of the total organic carbon in DOC and POC is based on the empirical selection of a particles **cut-off of 0.45 µm**: DOC < 0.45 µm, POC > 0.45 µm.



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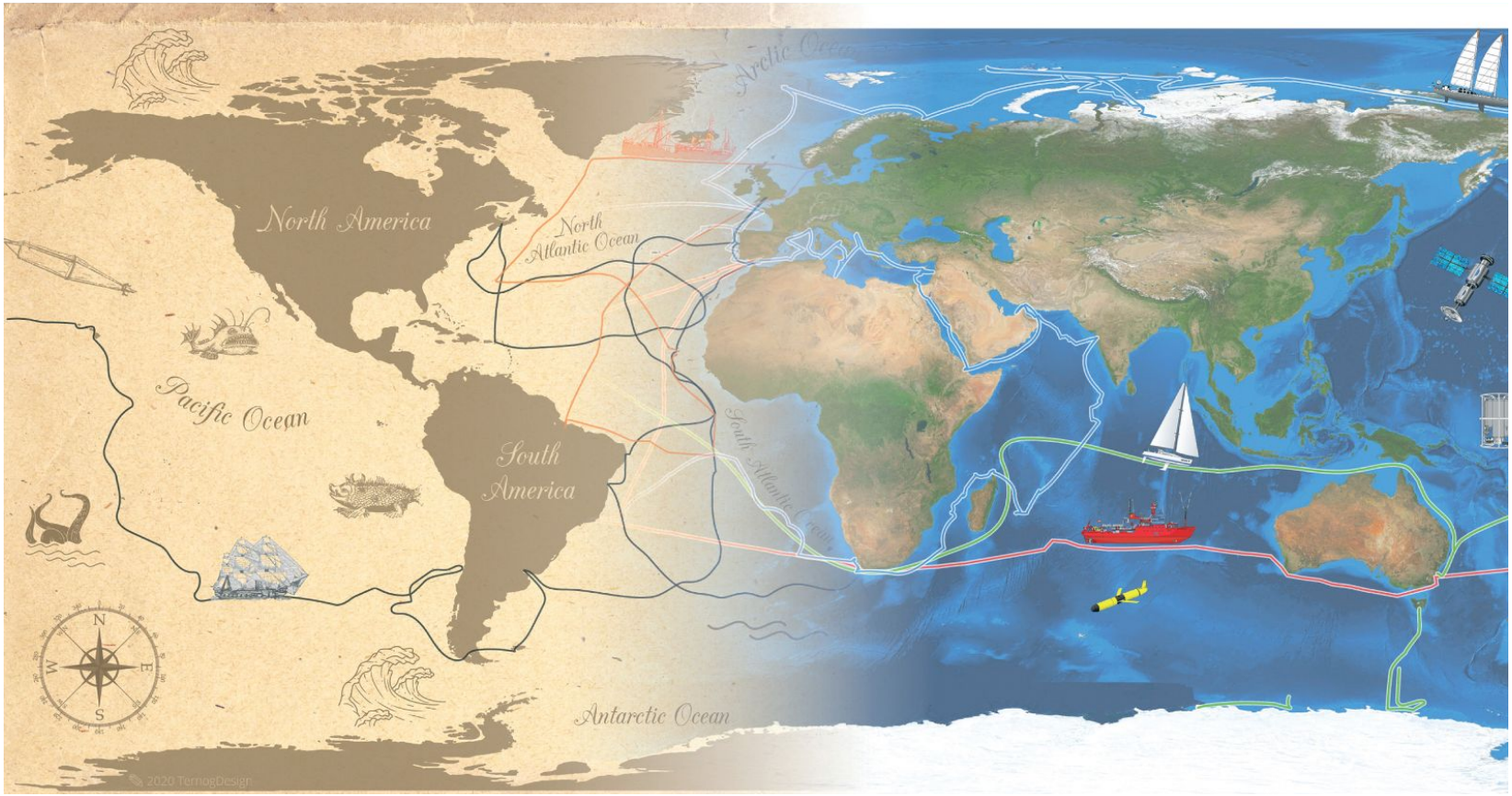
Primary production: POM and DOM

In general, the total organic matter distributed in an environment is referred as Dissolved and Particulate Organic Matter (DOM and POM). Each of the pool is constituted by the main organic nutrients: Carbon (DOC and POC) Nitrogen (DON and PON) and Phosphorus (DOP and POP).
DOM and POM, while size partitioned, are a continuum in nature. The main and practical difference resides in their sinking behavior. POM, constituted by cells, fecal pellets and cell debris, sinks efficiently, although different particles have different velocities. POM constitute one of the main export to the deep-sea floor. DOM on the other hands, constituted by DNA and small soluble molecules, viruses and picoplankton, does not sink.



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10.1038/nrmicro3326



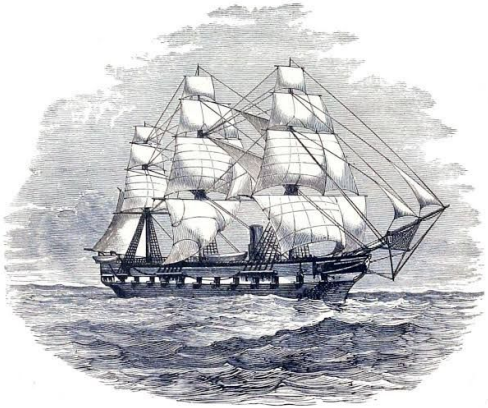
Oceanographic expeditions routes (1872-2013)

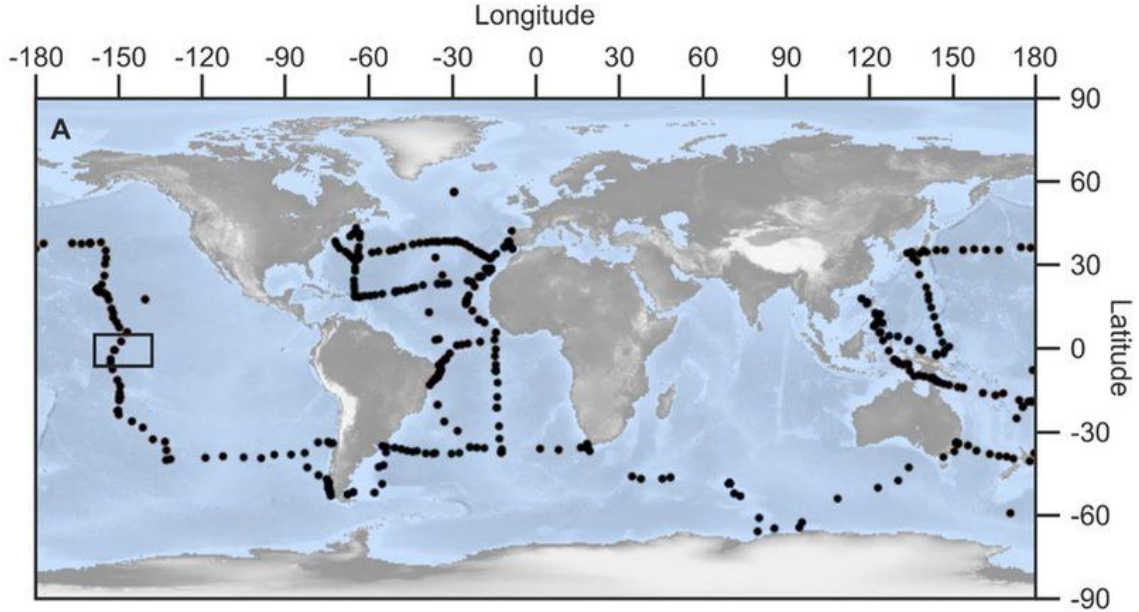
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10.1093/plankt/fbaa049

The Challenger Expedition (1872–1876)

The Challenger Expedition (1872–1876) was a pioneering global scientific voyage by the Royal Navy ship HMS Challenger that laid the foundation for modern oceanography. Nearly 69,000 nautical miles (~127,000 km) circumnavigating the globe across the Atlantic, Southern, Pacific, and Indian oceans. 362 observation stations established, 4,700+ new species found, discovered the Challenger Deep in the Mariana Trench, etc..





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DOI: 10.1038/s41598-020-58501-w

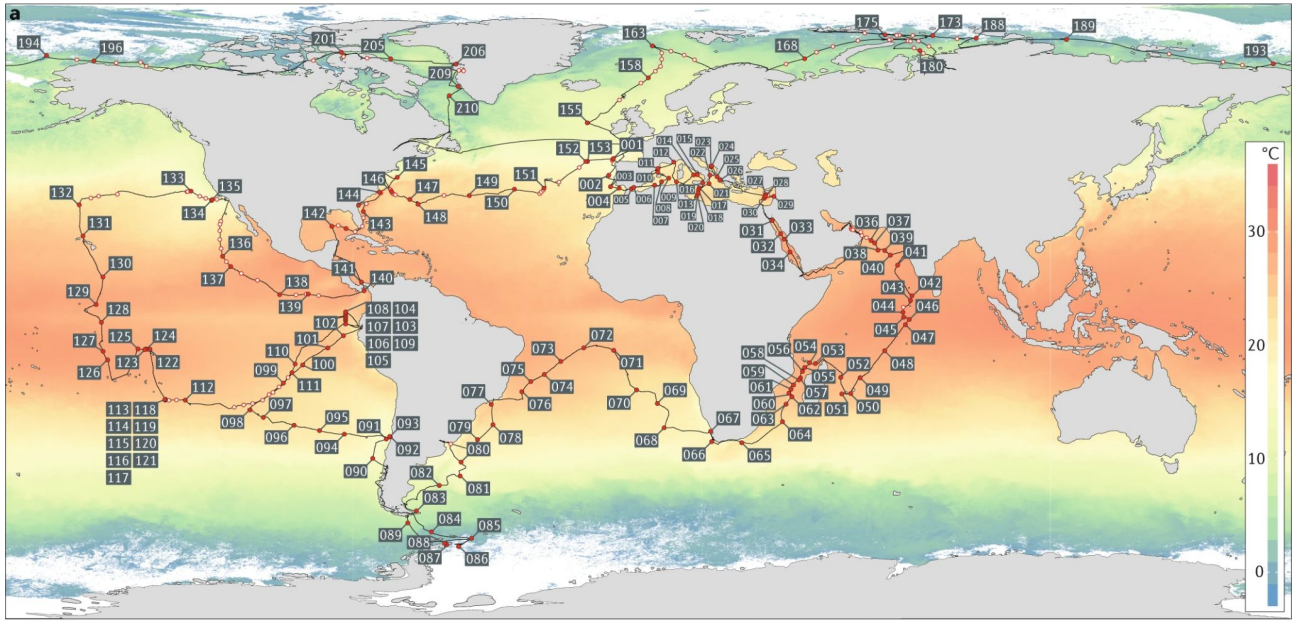
Tara Ocean

Tara Ocean is an international, multidisciplinary project to assess the complexity of ocean life across comprehensive taxonomic and spatial scales. Using a modified sailing boat, they sampled plankton at 210 globally distributed sites.



Tara Ocean

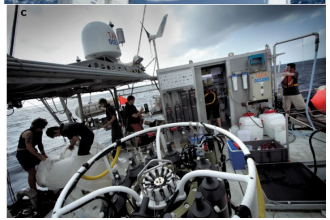
Tara Ocean is an international, multidisciplinary project to assess the complexity of ocean life across comprehensive taxonomic and spatial scales. Using a modified sailing boat, they sampled plankton at 210 globally distributed sites.



***Tara* Oceans (2009–2013)**

- 140,000 km sailed
- >35,000 plankton samples collected
- 210 sampling stations
- >60 terabases of DNA and RNA sequenced
- ~7 million images captured
- 120 crew members and scientists on-board
- 52 stopovers in 37 countries
- 35,000 schoolchildren on board at stopovers





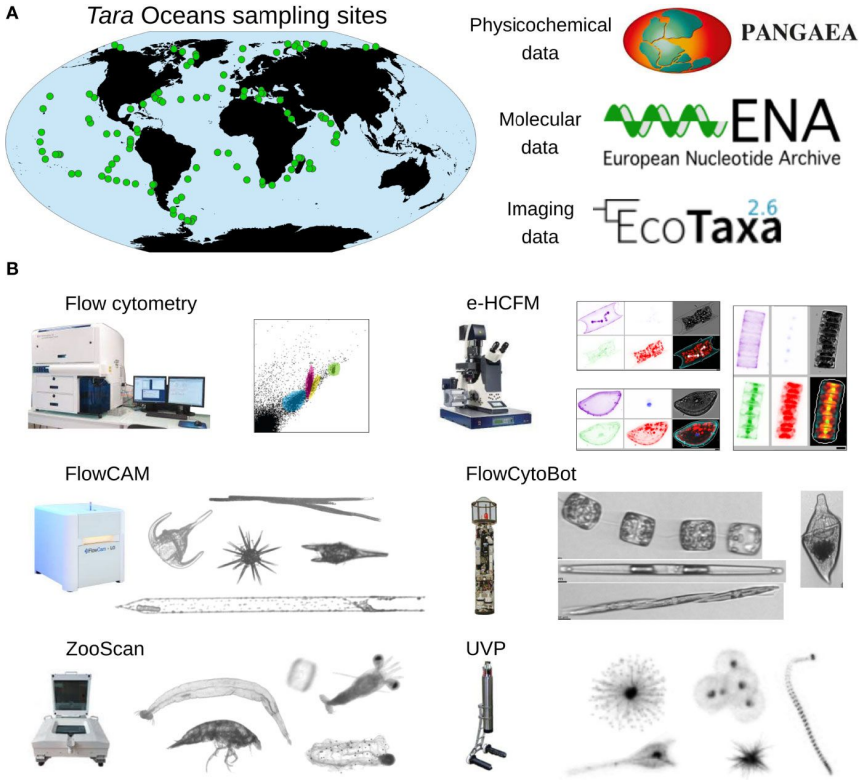
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DOI: 10.1038/s41579-020-0364-5

Tara Ocean

Tara Oceans expeditions (2009-2013) collected > 35000 plankton samples from 210 sampling sites of the upper ocean. High-throughput imaging and omic methods were used for generating physicochemical contextual data as well as > 60 terabases of DNA and RNA sequences and ~7 million images.

Tara Ocean project tried to answer very basic questions:
Which plankton group is the most abundant?
Which is the most diverse?

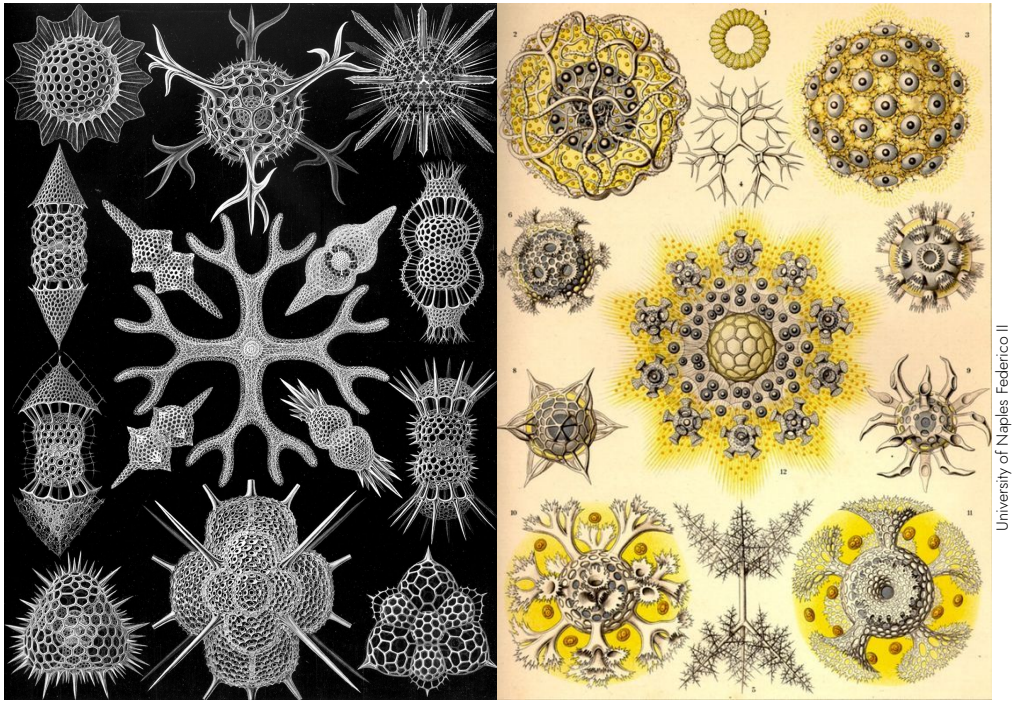


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DOI: 10.1038/s41579-020-0364-5

Tara Ocean: Uncovering Marine Biodiversity

Using data from a worldwide in situ imaging survey of plankton larger than 600 µm, they reported that the most abundant part of the biomass of this size fraction consists of giant protists belonging to the Rhizaria, a super-group of mostly fragile unicellular marine organisms that includes the taxa Phaeodaria and Radiolaria (for example, orders Collodaria and Acantharia). Globally, rhizarians were estimated to represent a standing stock of carbon, equivalent to 5.2% of the total oceanic biota carbon reservoir.



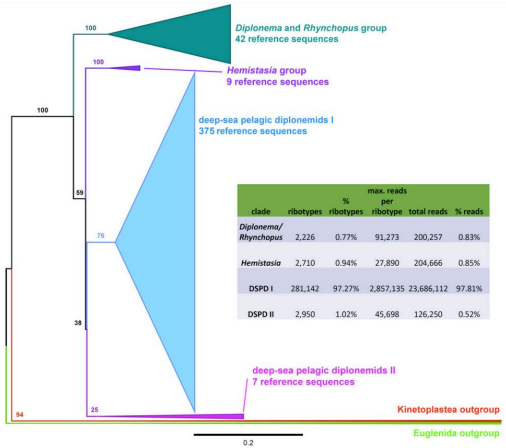
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10.1038/nature17652

Tara Ocean: Uncovering Marine Biodiversity

Tara Oceans metabarcoding study revealed that the **most diverse** part of the plankton in the upper sunlit layer of the ocean is represented by an understudied group of heterotrophic excavate flagellates called **Diplonemids**. They analyzed the diversity and distribution patterns of diplonemid populations on a set of 18S rDNA metabarcodes amplified from 850 size-fractionated plankton communities sampled across 123 globally distributed locations.
They reported diplonemids separation into four major clades, with the vast majority falling into the deep-sea pelagic diplonemid clade.
The diversity of diplonemids is highly stratified according to depth, with just 1883 OTUs (4.2%) shared across all three depth zones.





clade	ribotypes	% ribotypes	max. reads per ribotype	total reads	% reads
Diplonema/ Rhynchopus	2,226	0.77%	91,273	200,257	0.83%
Hemistasia	2,710	0.94%	27,890	204,666	0.85%
DSPD I	281,142	97.27%	2,857,135	23,686,112	97.81%
DSPD II	2,950	1.02%	45,698	126,250	0.52%

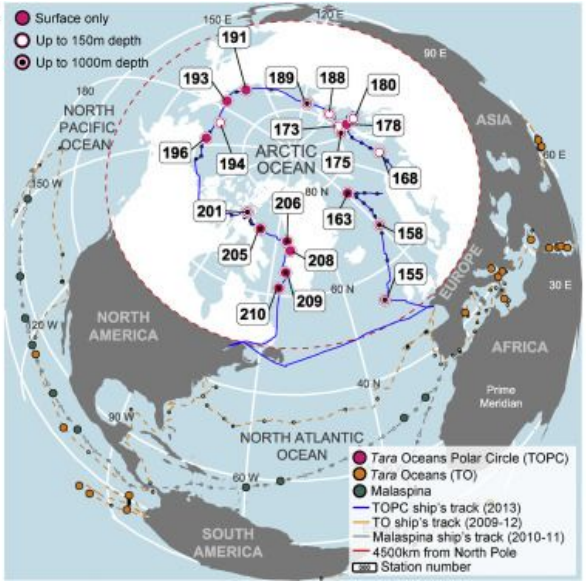


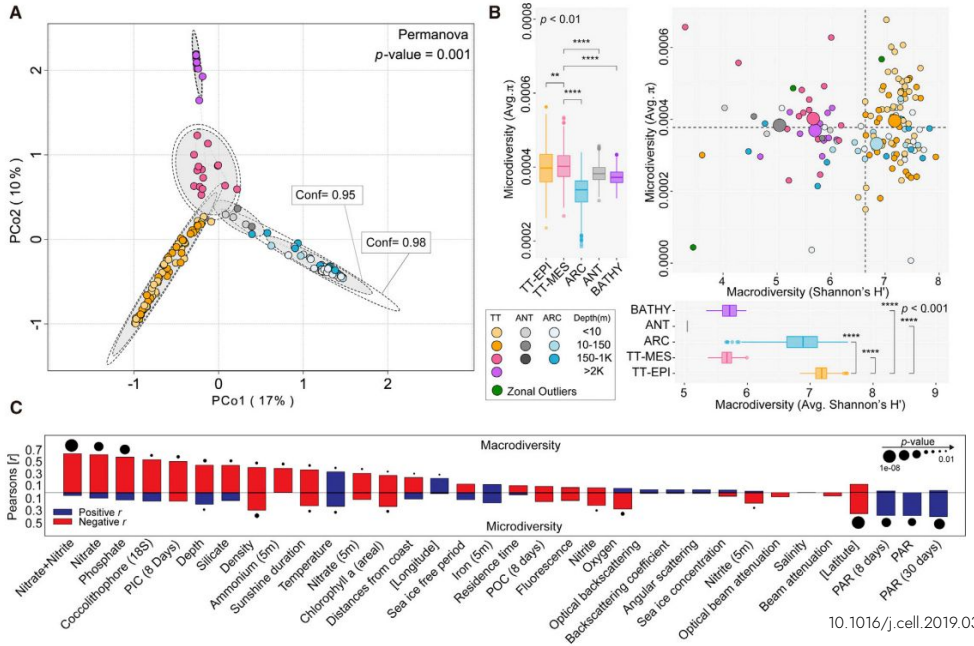
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10.1016/j.cub.2016.09.013

Tara Ocean: Mapping The Ocean's Viral Diversity

Meta-community analyses revealed five ecological zones throughout the global ocean, which are Arctic (**ARC**), Antarctic (**ANT**), bathypelagic (**BATHY**), temperate-tropical epipelagic (**TT-EPI**), and mesopelagic (**TT-MES**). Across the zones, local and global patterns and drivers in viral community diversity were established for both macrodiversity (inter- population diversity) and microdiversity (intra-population genetic variation).

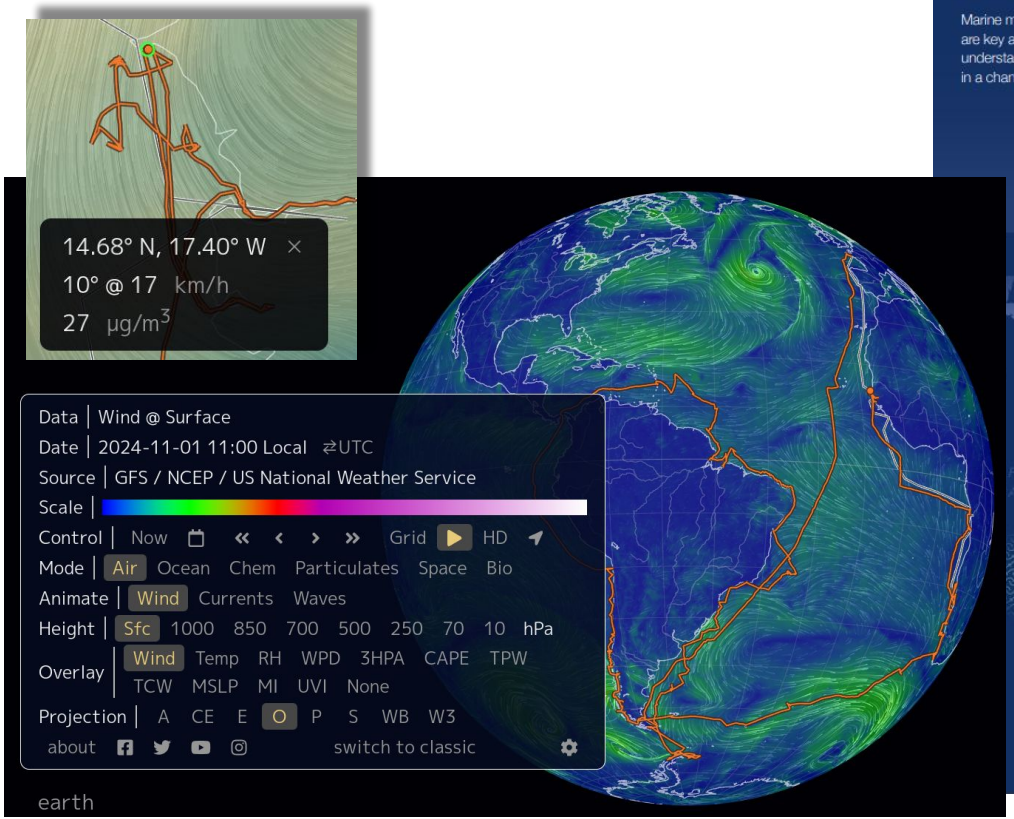




10.1016/j.cell.2019.03.040

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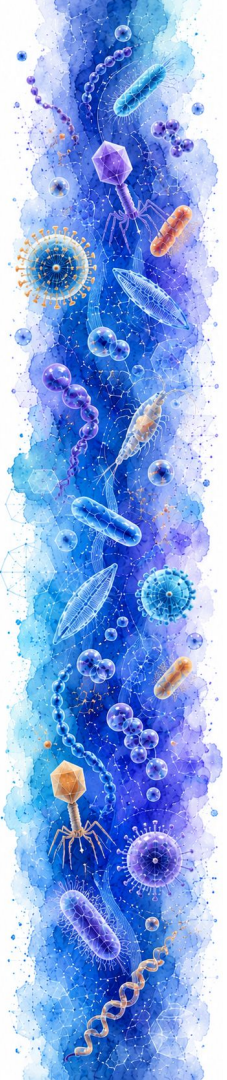
Tara Ocean: Marine Ecosystem Responses To Climate Changes And Pollution





https://fondationtaraocean.org/en/research-fields/marine-biodiversity-pollution/

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This Week Read

Sunagawa S, Acinas SG, Bork P, Bowler C; Tara Oceans Coordinators; Eveillard D, Gorsky G, Guidi L, Iudicone D, Karsenti E, Lombard F, Ogata H, Pesant S, Sullivan MB, Wincker P, de Vargas C. Tara Oceans: towards global ocean ecosystems biology. Nat Rev Microbiol. 2020 Aug;18(8):428-445. doi: 10.1038/s41579-020-0364-5.

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